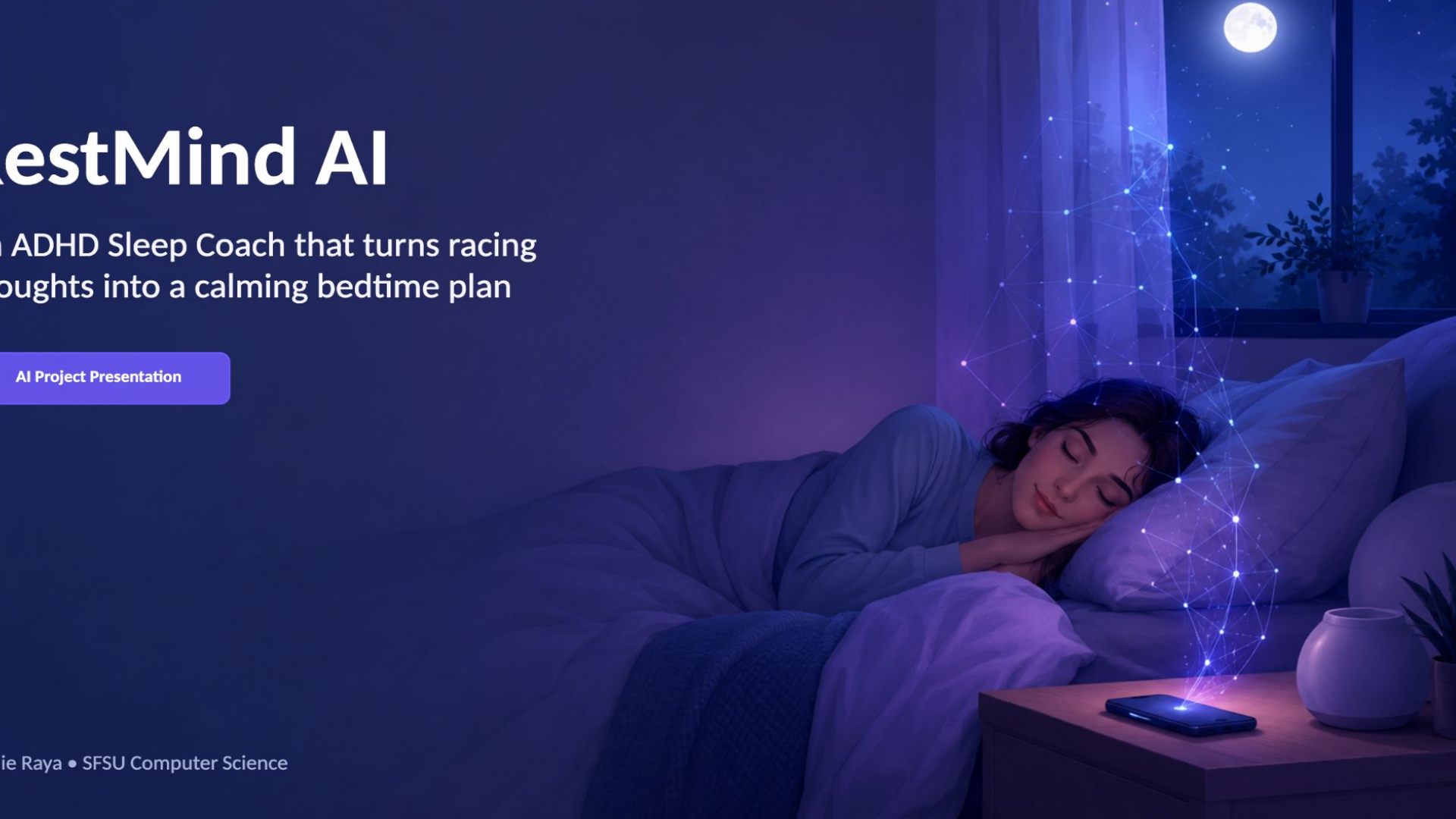


RestMind AI

ADHD Sleep Coach that turns racing
thoughts into a calming bedtime plan

AI Project Presentation



The Problem

ADHD can make bedtime feel like the brain is finally “wide awake.”

Why sleep becomes difficult

- Racing thoughts and unfinished tasks show up at night
- Phone scrolling becomes a quick dopamine loop
- A routine can be hard to start without structure
- Generic sleep advice does not adapt to the user

“I forgot...”

“What if...”

“One more video...”



Goal: calm the loop before sleep

Project Goal

A supportive AI assistant that helps users prepare for sleep, not diagnose or replace care.

RestMind AI helps the user:

1. Empty the mind

A fast “brain dump” turns random bedtime thoughts into organized tomorrow tasks, worries, or reminders.

2. Choose a calming action

The AI recommends a short, realistic strategy based on the user’s current mood and energy level.

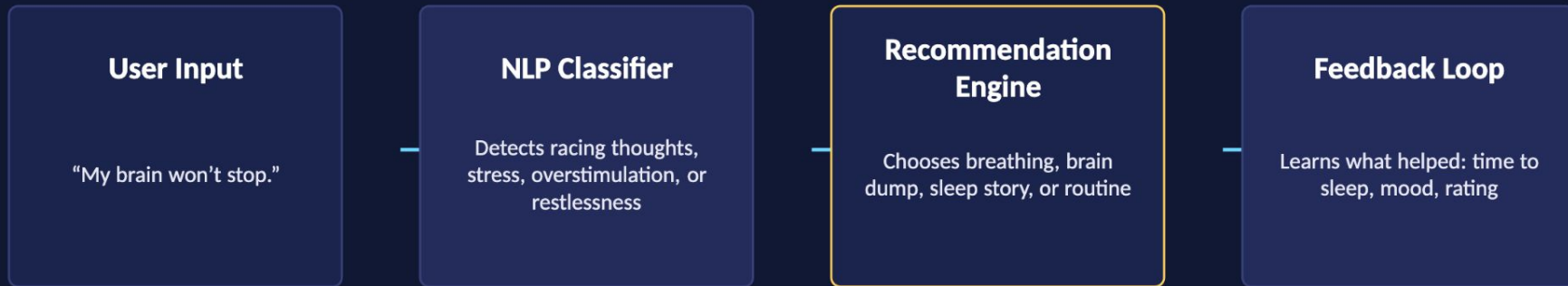
3. Build a routine

Over time, the assistant learns which wind-down steps actually help the user fall asleep faster.

Design promise: gentle, short, personalized, and nonjudgmental.

How the AI Model Works

Input → understand the sleep problem → recommend a calming response → learn from feedback.



Model type: personalized natural language AI + recommendation system

Core Features

The app gives the user one small action at a time instead of overwhelming them.



Brain Dump Organizer

Sort thoughts into tomorrow tasks, worries, reminders, and “let go for tonight.”



Wind-Down Coach

Suggests a short routine: dim lights, water, breathing, stretch, or audio.



Sleep Story Generator

Creates calm stories based on the user's interests, without stimulating plots.



Sleep Pattern Tracker

Learns what helps over time using ratings, bedtime, wake time, and energy.

Output stays short: one prompt, one action, one next step.

Sample User Flow

A simple example of how the assistant could respond at bedtime.

User: "I'm tired but I keep thinking about everything I need to do tomorrow."

AI detects: racing thoughts + task anxiety

Confidence: high

AI response:

"Let's do a 3-minute brain dump. I'll sort your thoughts into: do tomorrow, not urgent, and worry-only. Then we'll do one breathing exercise."

3-minute brain dump

Breathing exercise

This keeps the interaction short, supportive, and easy to follow.

AI Techniques + Tech Stack

A realistic build plan for a student project or portfolio demo.

AI Techniques

- Text classification for sleep-state detection
- Prompted LLM responses for calming stories and guidance
- Recommendation rules based on mood, energy, and history
- Optional GraphRAG: connect routines, triggers, and strategies as a knowledge graph

Tech Stack

Frontend: React or HTML/CSS

Backend: Python + FastAPI

Model: OpenAI / Groq / local LLM

Data: JSON or SQLite for user history

Deploy: GitHub Pages + Render

Inspired by the workshop approach: define the idea, prompt clearly, generate, review, refine, and deploy.

Ethics + Safety

Because sleep and ADHD are connected to mental wellness, the app must be careful.

Not medical diagnosis

The assistant should clearly state that it is a wellness support tool, not a doctor or therapist.

Privacy first

Sleep notes may include personal worries, so data should be stored securely and minimized.

Crisis-aware limits

If the user mentions self-harm or severe distress, the app should provide urgent support resources.

No shame design

Language should avoid guilt. The assistant should encourage restarting gently after bad nights.

Low stimulation

Avoid bright visuals, endless chats, or exciting stories that keep users awake.

Demo Plan

What I would show in a final class presentation.



Success metric: user completes a routine and reports feeling calmer before sleep.

Final Vision

RestMind AI is not just a sleep app. It is a supportive AI companion that helps people with ADHD calm their thoughts, build healthier routines, and feel more in control at night.

Thank you

Leslie Raya

